

Neil Kale

nkale@cs.cmu.edu | (508) 936-3455 | neilkale.github.io | www.linkedin.com/in/neil-kale/

Education

Carnegie Mellon University (CMU)
Master of Science in Machine Learning
GPA 4.24/4.0

Pittsburgh, PA
December 2025

Advisors Prof. Virginia Smith, Prof. Aditi Raghunathan

Committee Service MSML Admissions Committee, MSML Social Committee, SCS Master's Advisory Committee

Worcester Polytechnic Institute (WPI)

Worcester, MA

Bachelor of Science in Computer Science, **2nd Major** in Mathematics, **Minor** in Robotics Engineering

May 2024

GPA 4.0/4.0

Awards and Activities Graduation with High Distinction, National Academic Achievement (Upsilon Pi Epsilon), Meritorious Senior (Pi Mu Epsilon), Men's Varsity Rowing

Selected Publications

Multimodal AI

[1] Wu, C. H.*, **Kale, N.***, & Raghunathan, A. Why foundation models struggle with cross-modal context. In *2025 Conference on Language Modeling (CoLM)*.

[2] **Kale, N.**, Zhang, C. B. C., Zhu, K., Aich, A., Rodriguez, P., Team, S. R., ... & Wang, Z. (2025). Reliable Weak-to-Strong Monitoring of LLM Agents. *arXiv preprint arXiv:2508.19461*. Under review at ICLR 2026.

Privacy and Auditing

[3] **Kale, N.**, ..., & Smith, V. (2025). Position: Child Safety Necessitates New Approaches to AI Safety.

[4] Thaker, P., Hu, S., **Kale, N.**, Maurya, Y., Wu, Z. S., & Smith, V. (2025, April). Position: LLM unlearning benchmarks are weak measures of progress. In *2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)* (pp. 520-533). IEEE.

[5] Thaker, P.*, **Kale, N.***, Wu, Z. S., & Smith, V. (2025). Membership Inference Attacks for Unseen Classes. *arXiv preprint arXiv:2506.06488*. Under review at ICLR 2026.

[6] Hu, S., **Kale, N.**, Thaker, P., Fu, Y., Wu, S., & Smith, V. (2025). BLUR: A Benchmark for LLM Unlearning Robust to Forget-Retain Overlap. *arXiv preprint arXiv:2506.15699*.

Healthcare

[7] Gjuka, D., Adib, E., ..., **Kale, N.**, ..., Kwiatkowski, D., & Stone, E. (2023). Enzyme-mediated depletion of methylthioadenosine restores T cell function in MTAP-deficient tumors and reverses immunotherapy resistance. *Cancer cell*, 41(10), 1774-1787

Industry Experience

Scale AI (Hosted by **Christina Knight**)

October 2025 - January 2026

Security and Policy Research Lab Fellow

Engaging with global policymakers to solve real-world problems in agent robustness, AI control, and frontier risk evaluation.

Scale AI (Hosted by **Sail Wang & Julian Michael**)

May 2025 - August 2025

SEAL Research Scientist Intern

Developed novel chain-of-thought monitoring techniques to oversee superhuman agents with weaker LLM monitors.

GoDaddy Inc (Hosted by **Harsh Nilesh Pathak**)

Machine Learning Scientist Intern

May 2024 - August 2024

Built GoDaddy's first agentic pipeline for CRM and prototyped an early agentic chatbot.

Harvard Medical School (Hosted by **David Kwiatkowski**)

May 2021 – August 2023

Research Assistant

Analyzed 1,200 bladder cancer genomes/transcriptomes as the only computer scientist on a 20-person research effort over 2 years.

Teaching

Teaching Assistant for 10701 Introduction to Machine Learning (CMU), CS541 Deep Learning (WPI)

Relevant Coursework

Graduate 10715 Introduction to Machine Learning (A+), 36700 Probability and Statistics (A), 15789 Foundations of Modern Machine Learning (A+), 10708 Probabilistic Graphical Models (A+), 10725 Convex Optimization (A), 10703 Deep Reinforcement Learning (ongoing), 10718 Machine Learning in Practice (ongoing)

Undergraduate CS534 Artificial Intelligence (A), CS541 Deep Learning (A), MA4635 Statistical Learning (A)